iii. Body short circuit test or Ground fault test

The armature to be tested is placed on the growler as shown in the figure 10.9 and AC supply is given to the growler. Now voltage is induced in the armature winding according to the transformer action. And it can be measured by AC milli voltmeter. The two leads of the milli voltmeter is connected between the commutator segment and the shaft of the armature.

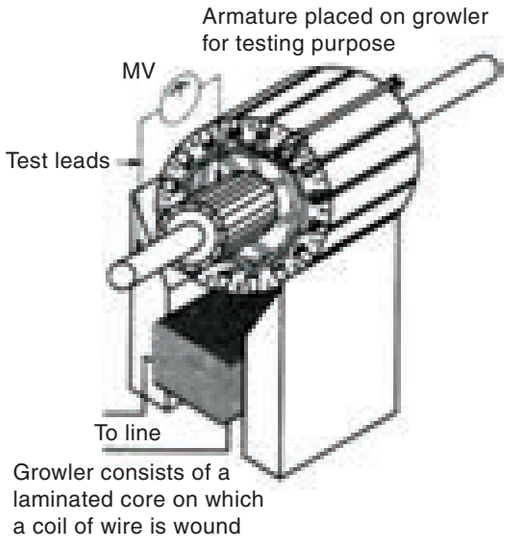


Fig. 10.9 Earth fault test

The pointer of the mili voltmeter must deflect to any value when we made test on all commutator segments. If the pointer does not deflect, the coil connected with the particular segment is subjected to contact with iron parts of the armature.

iv. Drop test

The most accurate method of testing the armature for open circuit, short circuit, ground fault, and reversal of connection can be done by the Drop test. Connect a low voltage DC supply across the commutator segment at distance of pole pitch.

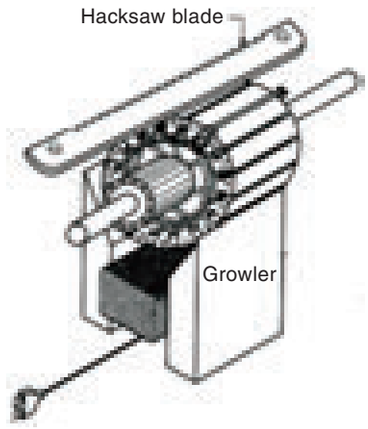


Fig. 10.8 Short circuit test

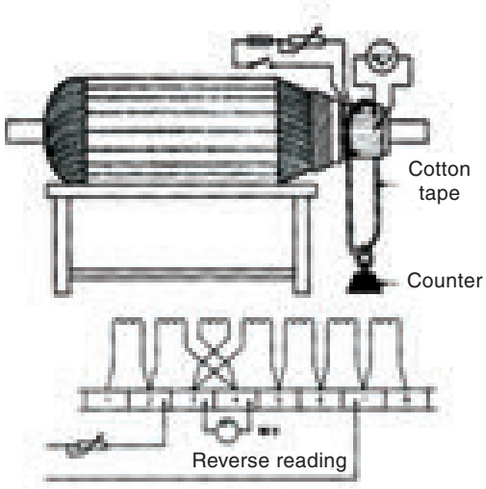


Fig. 10.10 Drop test

Insert a variable resistance in series as shown in the figure. For example, in two pole machines DC supply is given to the opposite brushes, and in four pole machine DC supply is given to the adjacent brushes.

Now place the DC millimeter leads with the adjacent bars and find the readings for all commutator segments. From the meter readings, we can conclude that

1. If all the readings are same, the windings is in good connection
2. If the meter reads zero or low voltage, the coil connected to the particular segment is short.

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